

★ WEEK 10 · PART 1 of 3

Hidden Markov Models: The Market Has Moods

A Discrete Hidden State That Switches Between Regimes

Markov chains · the transition matrix · emissions · market regimes

The Kalman filter tracked a continuous hidden state. Now the hidden state is discrete — a regime like calm or crisis — that the market switches between, revealed only through the returns it emits.

1

PART 1 · FROM CHAINS TO HMMs

A Hidden Regime, Emitting Returns

Markov chains and their one-step memory, the transition matrix, the leap from visible to hidden states, the four pieces of an HMM, and how regimes show up in markets.

From Pushkin to Speech Recognition — Part 1

How counting letters in a poem founded a field

In 1913, Andrey Markov analyzed the sequence of vowels and consonants in Pushkin's Eugene Onegin. He showed the letters were not independent — the next depended on the current. That was the first Markov chain.

The insight

- Successive letters carry dependence, not randomness.
- But the dependence reaches back only ONE step.
- Given the current letter, the past adds nothing.

Why it mattered

- It modeled sequences with minimal memory.
- Simple enough to compute, rich enough to be useful.
- It became the foundation of a huge field.

From Pushkin to Speech Recognition — Part 2

The hidden twist, and the lesson

Decades later, Leonard Baum and colleagues added a twist: what if the chain is HIDDEN, and you see only noisy signals of it? That hidden Markov model became the engine of speech recognition.

The twist

- The state (the spoken sound) is not directly observed.
- You see only the acoustic signal it produces.
- Infer the hidden sequence from the observations.

The lesson for us

- Markets hide their state too — calm or crisis.
- We see only the returns each regime emits.
- The same model infers the regime from the data.

From a Continuous State to a Discrete One

Where Week 9 left off

1

Kalman tracked a continuous state

Week 9 estimated a smoothly-evolving hidden state — a level, a drifting beta — with a linear-Gaussian model.

2

But some states are discrete

A market is arguably in one of a few regimes: calm or crisis, bull or bear. That state doesn't drift smoothly — it SWITCHES.

3

HMMs model the switching hidden state

A hidden Markov chain jumps between regimes, and each regime emits returns from its own distribution. Same spirit as Kalman, discrete state.

Part 1 in one line: a hidden regime switches over time and reveals itself through the returns.

Markov Chains: Memory of Just One Step

The future depends only on the present state

A Markov chain is a sequence of states where the next state depends only on the current one — not on the whole history. Once you know where you are, the past tells you nothing more.

The one-step memory

- The next state's probabilities depend only on the current state.
- The path taken to get here is irrelevant.
- This is the Markov property — memorylessness beyond one step.

Why it is powerful

- It makes long sequences tractable to model and compute.
- A few transition probabilities describe the whole process.
- It is simple, yet captures real persistence.

The Markov Property

One line that defines the whole idea

$$\text{Markov property: } P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_1) = P(X_{t+1} \mid X_t)$$

Reading it

- The probability of the next state, given all history...
- ...equals its probability given only the current state.
- The conditioning on the past collapses to the present.

The intuition

- The present state is a sufficient summary of the past.
- It carries everything relevant for the future.
- That is exactly what makes the chain memoryless.

The Transition Matrix

All the switching probabilities in one grid

$$A_{ij} = P(X_{t+1} = j \mid X_t = i), \quad \sum_j A_{ij} = 1 \quad (\text{each row sums to one})$$

What it is

- Entry (i, j) is the probability of moving from state i to state j.
- Each row is a probability distribution — it sums to one.
- It fully describes the chain's dynamics.

Reading the diagonal

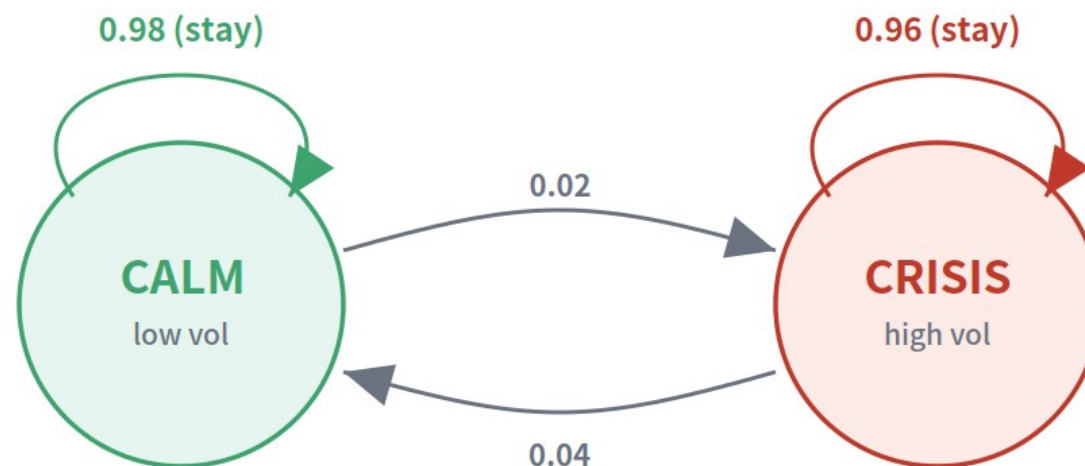
- Large diagonal entries mean the state tends to STAY.
- That persistence is called stickiness.
- Sticky regimes produce long calm and crisis spans.

A Two-State Markov Chain

The transition matrix, drawn. High stay-probabilities (the self-loops) make each regime persistent; the small cross-arrows are the occasional switches between calm and crisis.

A Two-State Markov Chain

The transition matrix, drawn: probabilities of staying vs switching.



Regimes are STICKY

High stay-probabilities mean each regime persists - so returns cluster into long calm and crisis spans.

Sticky Regimes and the Long Run

Persistence, and where the chain settles

Stickiness

- High stay-probabilities keep the chain in a regime.
- So states persist rather than flickering each step.
- This is why volatility clusters into regimes.

The stationary distribution

- Over the long run, the chain spends a fixed fraction of time in each state.
- That fraction is the stationary distribution.
- It tells you how common calm vs crisis is overall.

From Visible to Hidden

We don't get to see the state

In a plain Markov chain you observe the state directly. The crucial leap of an HMM: the state is HIDDEN. You see only observations it emits, and must infer the state behind them.

Plain Markov chain

- The state itself is observed at each step.
- You can read the transitions directly.
- Nothing is hidden.

Hidden Markov model

- The state is latent — never observed.
- Each state emits an observation you DO see.
- The whole challenge is inferring the hidden state.

The Pieces of an HMM

What fully defines the model

an HMM = (π, A, B) : initial π , transition A , emissions B

The three parts

- An initial distribution over the starting state.
- A transition matrix for switching between states.
- An emission model — each state's observation distribution.

Plus one choice

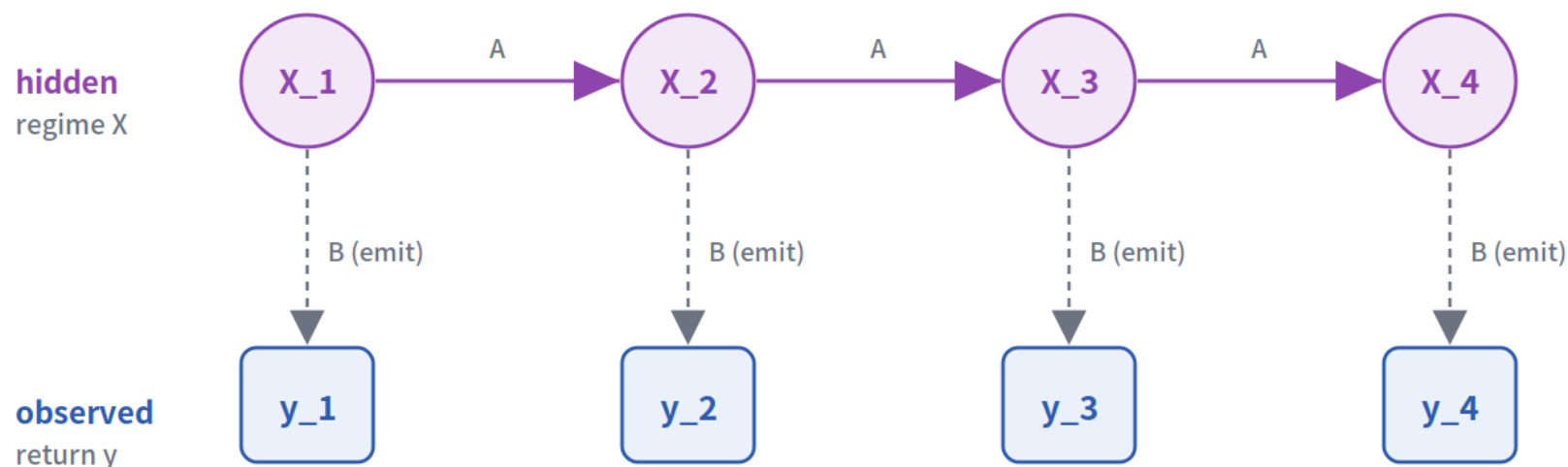
- The NUMBER of hidden states (e.g. two: calm, crisis).
- That count is yours to choose (Part 3).
- Together these fully specify the HMM.

The HMM Picture

A hidden regime chain switches by the transition A; each regime emits an observed return by the emission model B. We see only the bottom row and infer the hidden regimes above.

The HMM Picture

A hidden regime chain switches by A and emits the returns we observe by B.



The regime is never observed. Its distribution (calm vs crisis) shapes the returns - and we infer it from them.

Emissions: Each Regime Its Own Distribution

How a hidden state shows up in the data

$$\text{emission (Gaussian): } y_t \mid (X_t = i) \sim \mathcal{N}(\mu_i, \sigma_i^2)$$

The idea

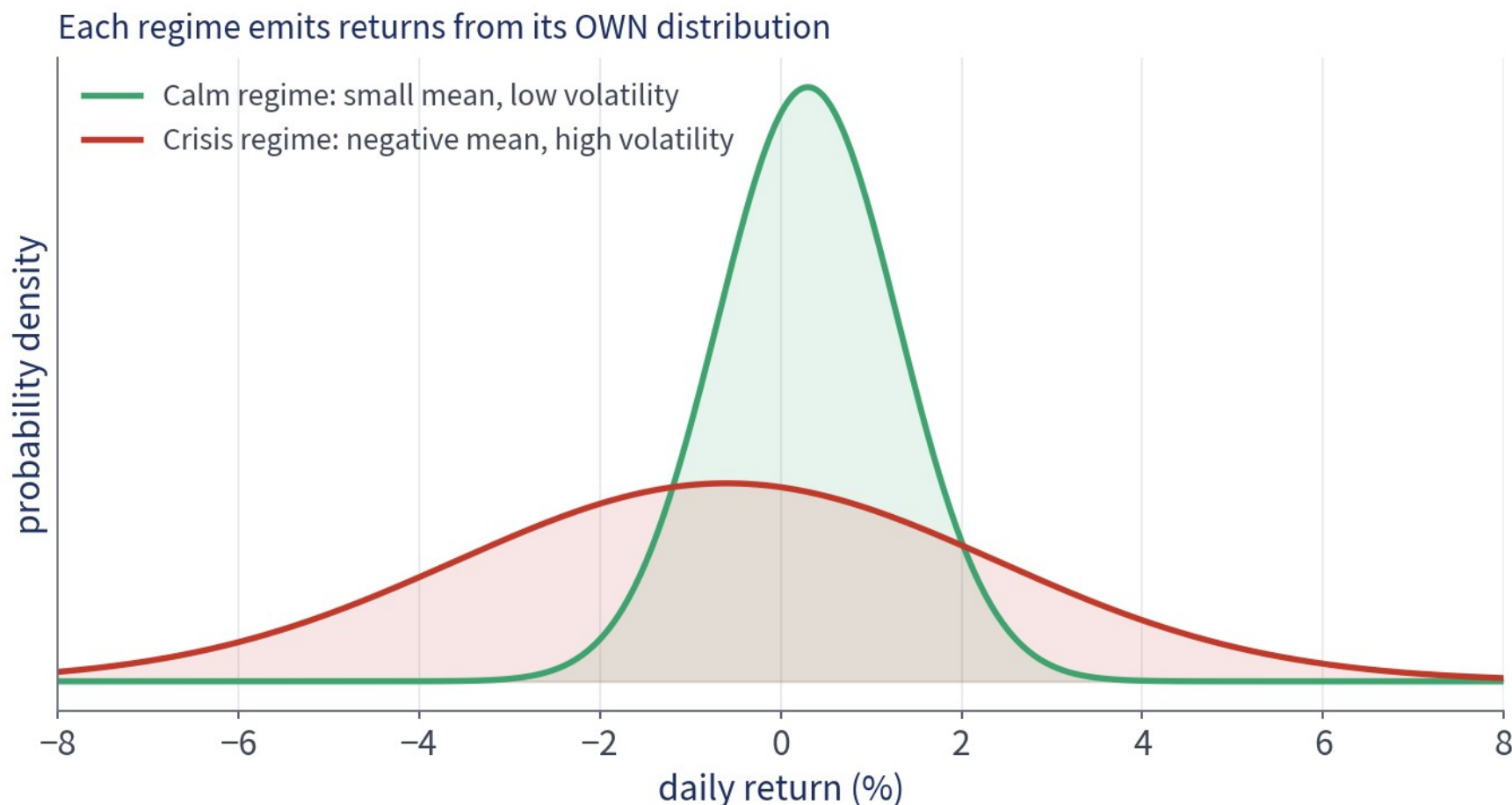
- Each regime emits observations from its own distribution.
- A calm regime: small mean, low variance returns.
- A crisis regime: negative mean, high variance returns.

Why this reveals the state

- Different regimes leave different fingerprints on the data.
- A run of large, wild returns hints at a crisis regime.
- The emission model is the bridge from state to observation.

Two Regimes, Two Distributions

The calm regime emits a tight distribution of small returns; the crisis regime emits a wide one with a negative mean. Which regime you're in shapes what returns you tend to see.



Regimes in Markets

The canonical financial use of HMMs

The natural financial reading: the market is in a hidden regime — calm or crisis, bull or bear — and returns are the emissions. The HMM infers which regime is active from the returns alone.

The regimes

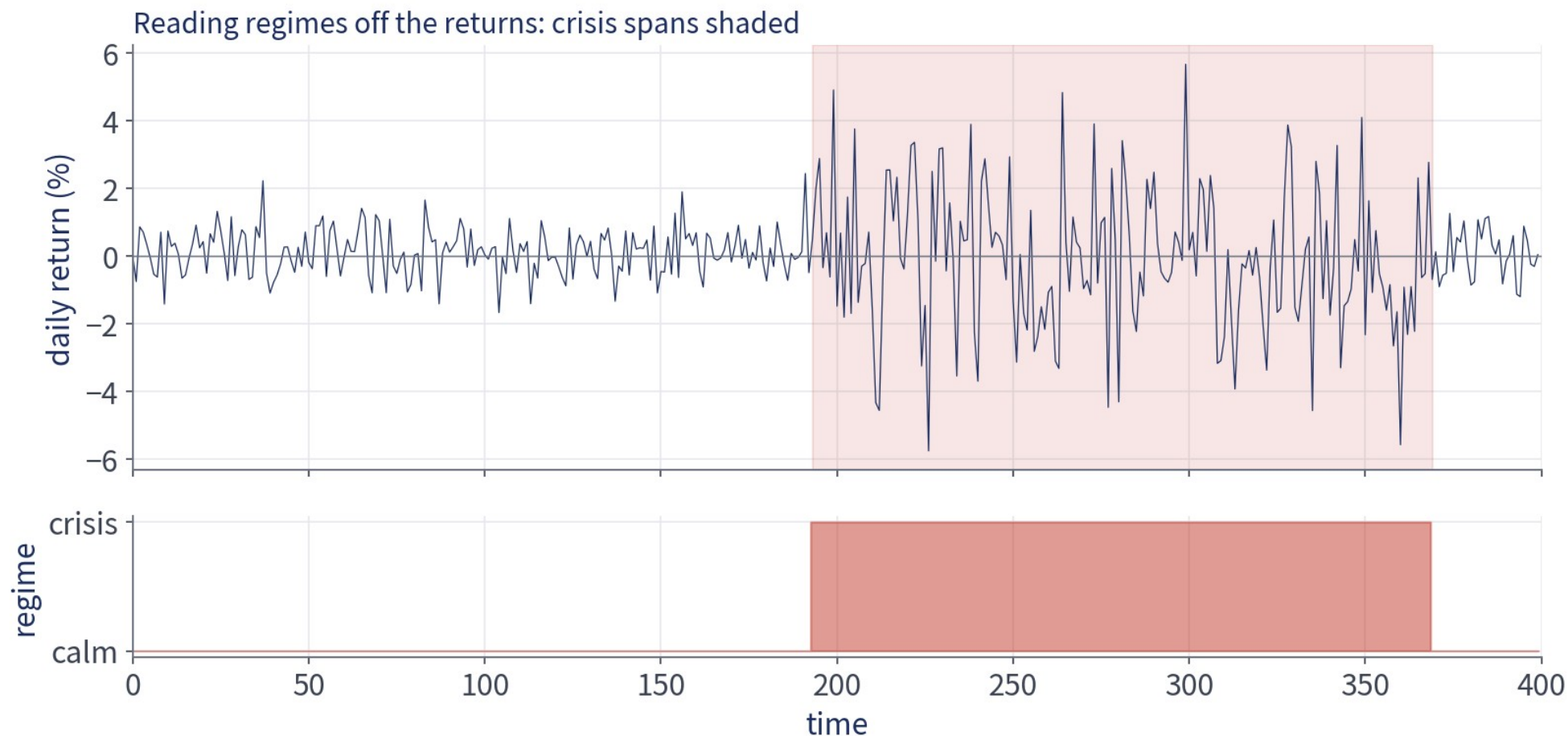
- Calm: modest positive drift, low volatility.
- Crisis: negative drift, high volatility, fat tails.
- Each is a state of the hidden chain.

What the HMM gives

- A probability of being in each regime, each day.
- A decoded regime path over history.
- An early read when the market shifts mood.

Reading Regimes Off the Returns

A return series with crisis spans shaded and the decoded regime below. Calm periods are quiet; crisis periods are volatile and clustered — exactly what a sticky, two-state HMM captures.



Why HMMs Fit Markets

Regimes match what returns actually do

What returns show

- Volatility clusters — quiet stretches and stormy stretches.
- Fat tails — occasional extreme moves.
- Persistence — turbulence tends to beget turbulence.

How HMMs capture it

- Sticky regimes produce the clustering directly.
- A high-volatility regime supplies the fat tails.
- A discrete switch matches sudden shifts in mood.

HMM vs Kalman Filter

Discrete state vs continuous state

Aspect	Kalman filter (Week 9)	HMM (Week 10)
Hidden state	continuous	discrete — a few regimes
State dynamics	linear-Gaussian transition	a transition matrix
Emissions	linear plus Gaussian noise	one distribution per regime
Inference	predict-update recursion	forward / Viterbi / Baum-Welch

Same core idea — a hidden state seen through noisy data — but the state is now discrete.

The Generative Story

How an HMM produces a return series

The recipe

- Pick a starting regime from the initial distribution.
- Emit a return from that regime's distribution.
- Switch regime by the transition matrix, then repeat.

Why it helps

- It tells you exactly what the model assumes.
- Inference simply reverses this story.
- You can simulate synthetic regime data to test methods.

What HMMs Assume

Two assumptions to keep in mind

$P(y_t \mid X_t)$ depends only on the current regime X_t (conditional independence)

The two assumptions

- Markov: the regime's next step depends only on the current regime.
- Conditional independence: each return depends only on its own regime.
- Given the regime, the observation carries no extra memory.

Why they matter

- They are what make the algorithms efficient (Part 2).
- They are approximations — real markets bend them.
- Knowing them tells you when an HMM may mislead.

What Parts 2 and 3 Add

From model to method to use

Part 2 — the algorithms

- Evaluation: the forward algorithm for the likelihood.
- Decoding: Viterbi for the most likely regime path.
- Learning: Baum-Welch to fit a model with hidden states.

Part 3 — in practice

- How many regimes to choose, and the pitfalls.
- Pairs trading Stage IV: trade only in favorable regimes.
- Cautions, and the bridge to continuous-time models.

Fit a 2-State Gaussian HMM

Direct Claude Code to detect calm and crisis

GOAL Fit a two-state Gaussian HMM to a return series and see the calm and crisis regimes it recovers.

[INPUT] Prompt · Claude Code

Write and run `hmm_regimes.py` that:

1. loads a daily return series (an index is fine);
2. fits a 2-state Gaussian HMM (e.g. `hmmlearn GaussianHMM` with 2 components);
3. prints each regime's mean and volatility, and the transition matrix (note the stickiness);
4. plots the returns with the decoded regimes shaded (calm vs crisis).

Tell me which state is 'crisis' (higher vol, negative mean) and how persistent each regime is.

How to work with it

- The crisis state has higher volatility.
- Diagonal transition entries should be near 1.
- Regimes should form long clustered spans.
- Keep it; Part 2 explains how it's fit.

Deliverable: `hmm_regimes.py` + regime parameters and a shaded return plot.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 In a Markov chain, what does the next state depend on?

Only the current state

Q2 What makes a Markov model 'hidden'?

The state is unobserved

Q3 How does a hidden regime reveal itself in the data?

Through its emissions

All three come from the Markov property, the hidden leap, and the emission model.

Three Misconceptions to Avoid

Where HMM intuition slips

Common myths

- "The regimes are labeled in the data."
- "An HMM needs long memory to model persistence."
- "The number of regimes is given by the data."

The reality

- Regimes are latent — the HMM infers them, unlabeled.
- One-step memory plus sticky transitions gives persistence.
- You CHOOSE the number of regimes; it's a modeling decision.

Three Things to Carry Forward

Part 1 in one screen

1

A Markov chain has one-step memory

The next state depends only on the current one, encoded in a transition matrix. Sticky (high-diagonal) chains persist, producing the clustering we see in markets.

2

An HMM hides that chain

The state is latent; we see only observations each regime emits. An HMM is an initial distribution, a transition matrix, and an emission model per state.

3

In markets, the state is the regime

Calm and crisis emit different return distributions. The HMM is the discrete-state cousin of the Kalman filter — same idea, a switching hidden state.

Next — Part 2: the three problems and their algorithms — forward, Viterbi, and Baum-Welch.